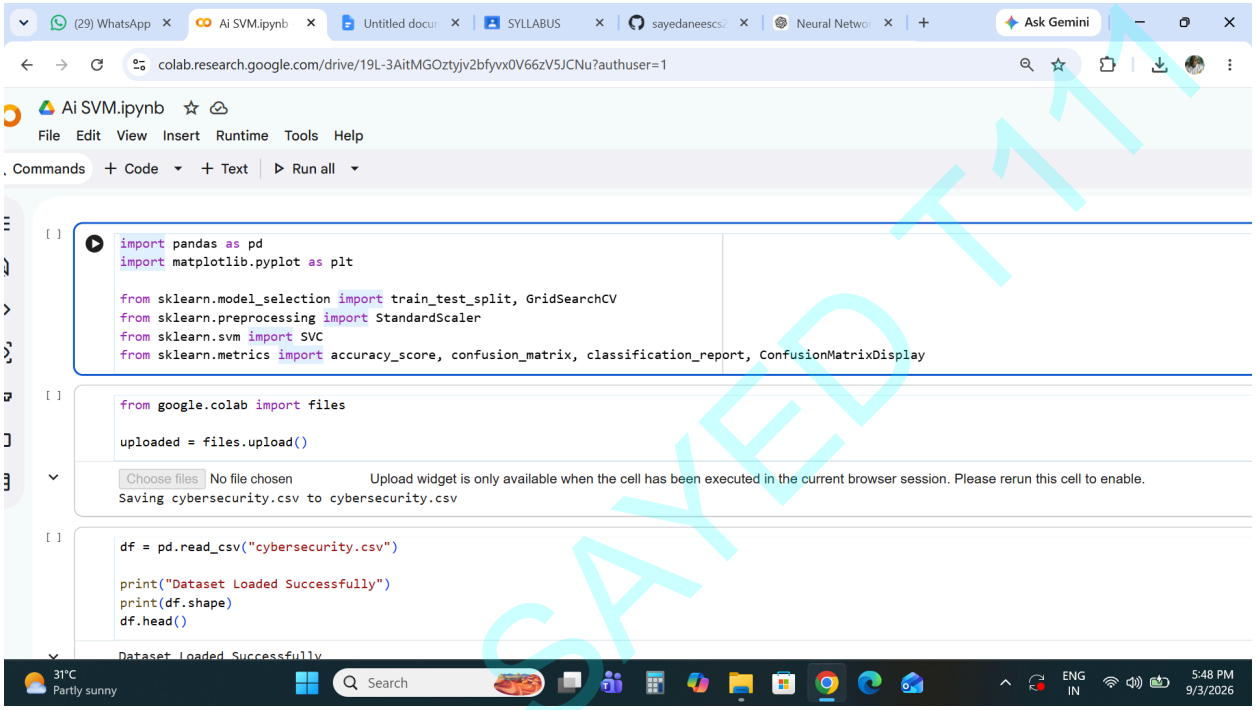
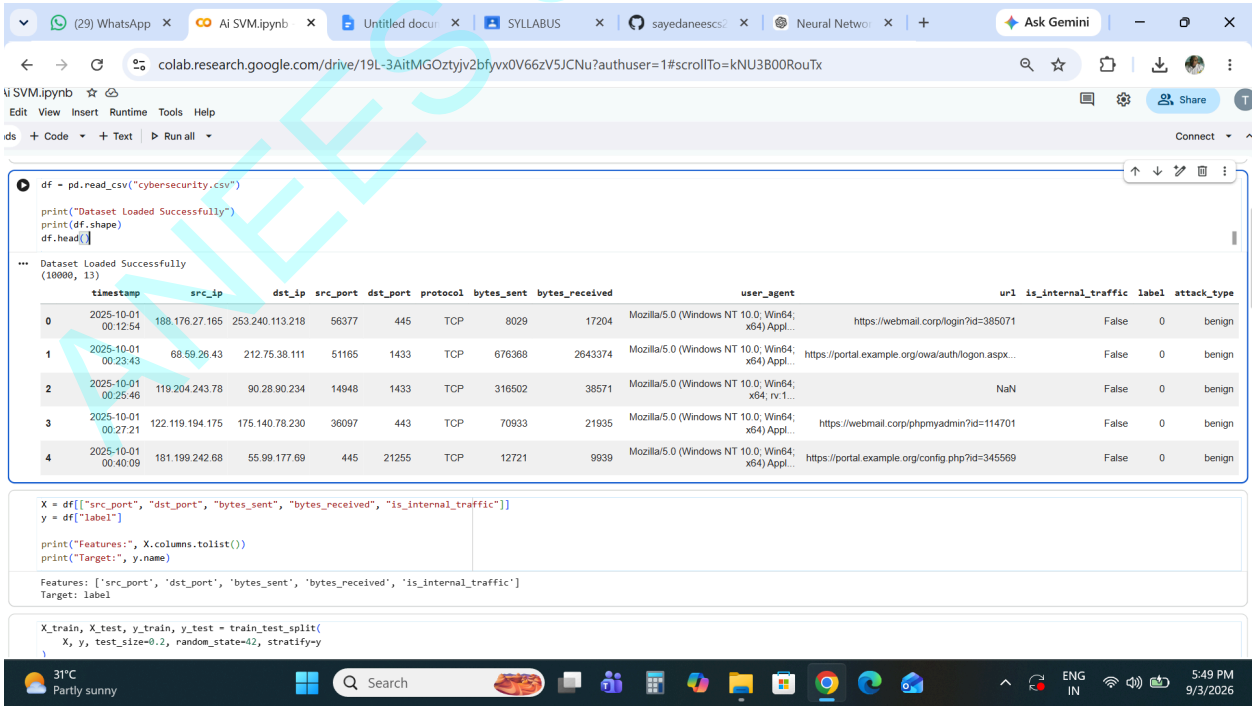
SHETH MVLU COLLEGE
SUBJECT:- AI

Practical 6

AIM:- Support Vector Machines (SVM)

- Implement the SVM algorithm for binary classification.
- Train an SVM model using a given dataset and optimize its parameters.
- Evaluate the performance of the SVM model on test data and analyze the results.





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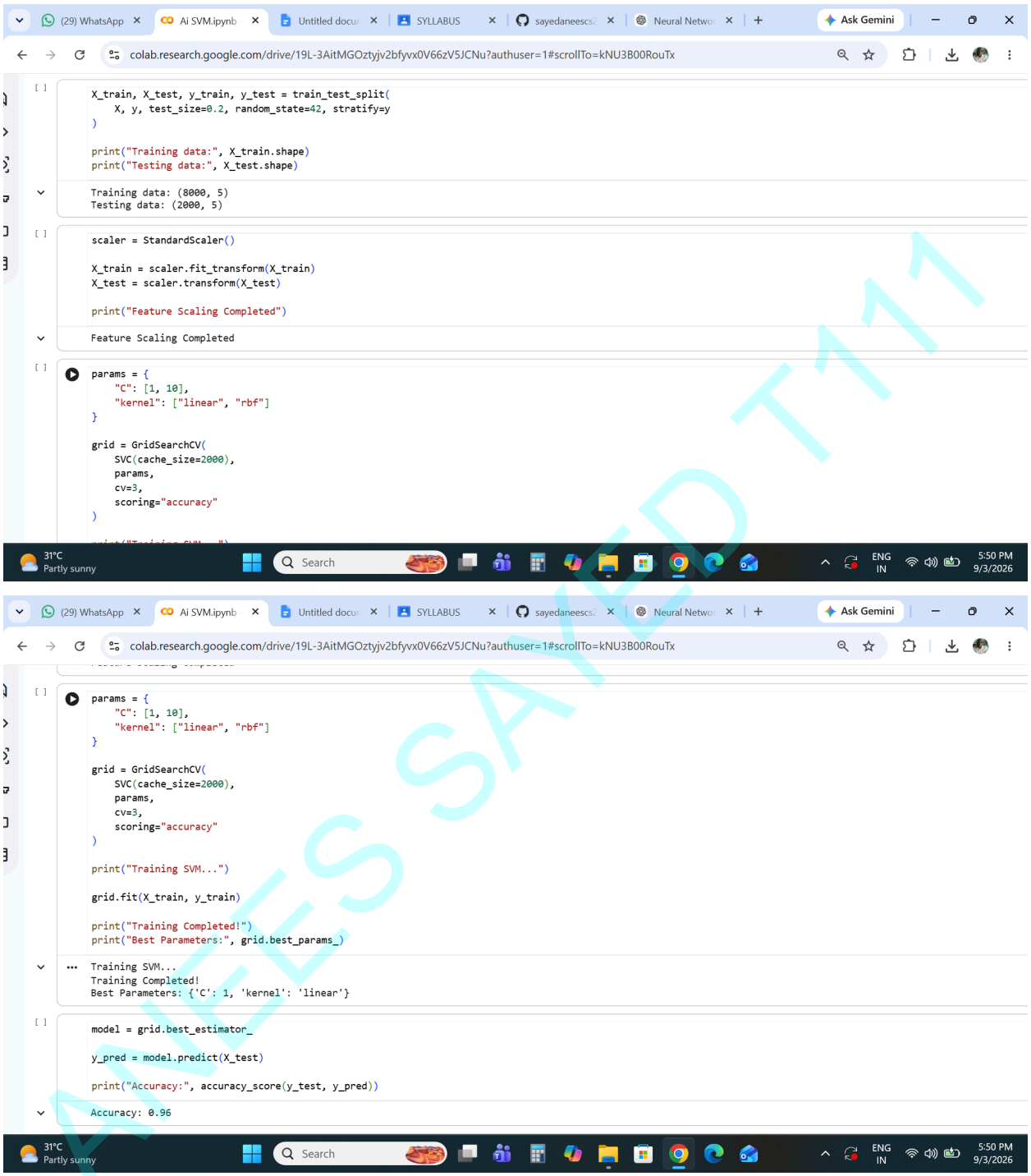

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

```
Training data: (8000, 5)
Testing data: (2000, 5)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

print("Feature Scaling Completed")
```

```
Feature Scaling Completed
```

```
params = {
    "C": [1, 10],
    "kernel": ["linear", "rbf"]
}

grid = GridSearchCV(
    SVC(cache_size=2000),
    params,
    cv=3,
    scoring="accuracy"
)

print("Training SVM...")

grid.fit(X_train, y_train)

print("Training Completed!")
print("Best Parameters:", grid.best_params_)
```

```
Training SVM...
Training Completed!
Best Parameters: {'C': 1, 'kernel': 'linear'}
```

```
model = grid.best_estimator_

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.96
```

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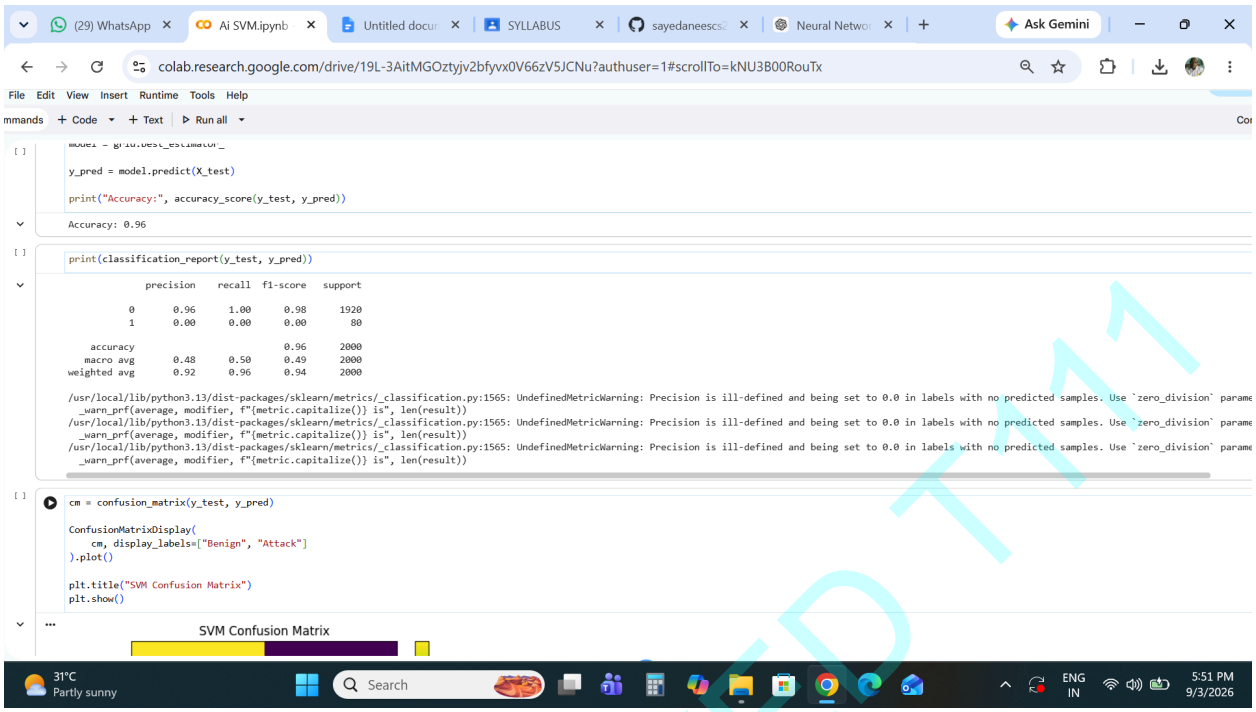

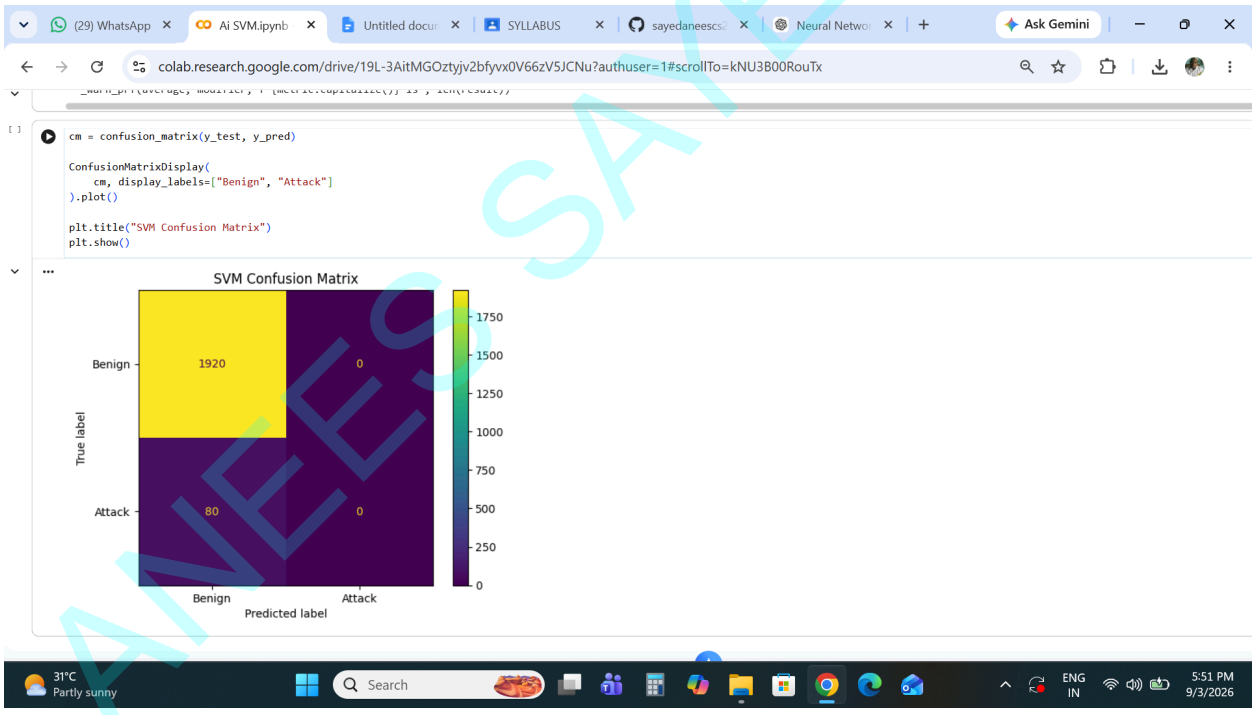

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